c. The fibril associated collagens connect two fibril forming collagens or a fibril forming collagen with other components of extra cellular matrix.

More than 1000 disease causing mutations have been identified both directly in collagen genes or in the genes of enzymes involved in the synthesis of collagen. The diseases associated with collagen malfunction are known as collagenopathies. Elhers-Danlos syndrome (EDS) is a prominent collagenopathy which is due to inherited mutations in collagen processing enzymes. Stretchy skin, loose joints and vascular problems are associated with EDS.



Figure 3.17 Stretchy skin in EDS syndrome

Osteogenesis Imperfecta (O.I) is another prominent collagenopathy characterized by brittle bones, hunch back, twisted spine and retarded wound healing. The mutations in glycine residues of collagen and thereby an improper triple helical structure leads to this disease.

3.7 Denaturation and Protein folding

Each protein has a unique three dimensional structure. Upon changes in various factors like temperature, pH, ionic strength or exposure to certain chemicals like urea lead to disruption of its three-dimensional structure and turn back into an unstructured string of amino acids. When a protein loses its higher-order structure, but not its primary structure, it is said to be denatured. Denatured proteins are not functional.

For some proteins, denaturation can be reversed. Since the primary structure of the polypeptide is still intact it may be able to re-fold into its original structure, if it is returned to its normal environment. Many proteins do not fold by themselves, but instead get assistance from other proteins like chaperons.

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